HERMES

# HERMES

## High-Frequency Emergency and Rural Multimedia Exchange System

Email, files, messages and voice over HF radio, where there is no internet

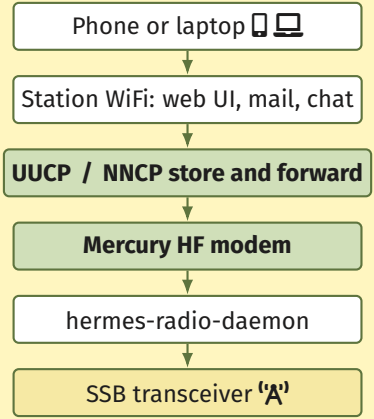
**Rhizomatica** • **free software** • **November 2026**

github.com/Rhizomatica • hermes.radio

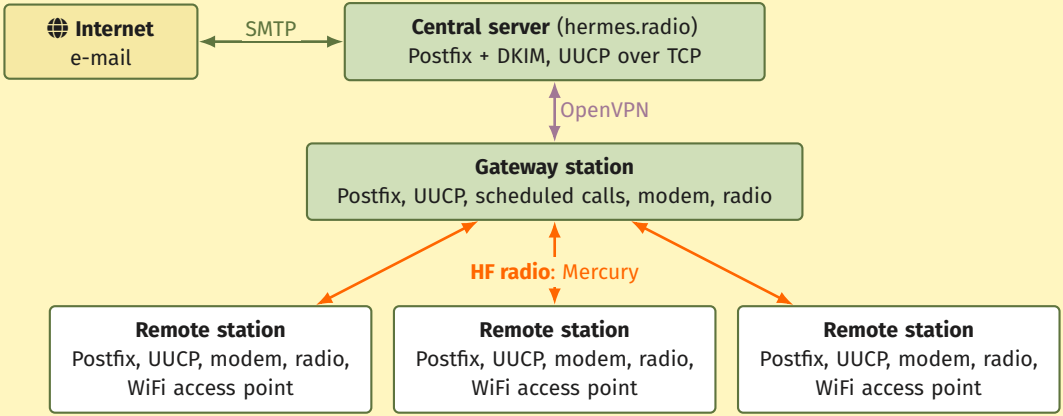


# Overview

- A **telecommunication system over HF radio** for communities beyond the reach of the internet and the phone network
- People use it as they use the internet: **email**, webmail and Delta Chat on their phone, over the station's **WiFi**
- The station carries the traffic over **HF** to a gateway, which has an uplink to the internet
- Also **voice**, digital and encrypted where that is permitted, **broadcast** files, and **positions** with an emergency call
- **Free software** from Rhizomatica, funded by ARDC and others; first deployed in the Amazon rainforest



# How a network fits together




The central server queues mail for each gateway. The gateway calls the central server over the VPN every few minutes, and its remote stations over HF on schedules set in its web interface.

Source: hermes-installer docs/README.md, docs/deploying-a-network.md

# A station

## Hardware

- A **Raspberry Pi** (1 to 16 GB) or another Debian computer
- An **SSB transceiver**: the **sBitx** or zBitx, driven directly, or any radio Hamlib controls (IC-7100, IC-7300, FT-710, TS-480, …)
- An antenna, and a WiFi access point for the users
- Optionally a GPS receiver

## Software

- Debian 13 and the HERMES package repository
- Postfix and Dovecot, UUCP (or NNCP), Mercury, **hermes-radio-daemon**
- A web interface for messages, users, the radio, the UUCP queue and schedules
- One installer for every station, from a **station file**; a USB installer image for the field

Source: hermes-installer docs/station-files.md; hermes-radio-daemon README.md

# Data over HF: store and forward, and broadcast

- **UUCP** carries mail and files over HF by default; it is robust, resumes interrupted transfers, and every station already knows it
- **NNCP**, encrypted and authenticated, is the alternative; a network uses one or the other on every station
- The link is **Mercury**: an ARQ session that delivers a file intact, or reports that it did not
- A station keeps mail while the band is closed and sends it at the next call

## Mercury, the link

Seven speed rungs, from MFSK below −10 dB SNR to 3.1 kbit/s; a receiver-driven, erasure-coded carousel; a VARA-compatible TNC. It has its own presentation.

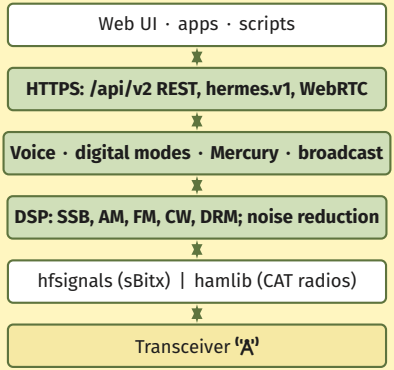
## Broadcast: one to many

Files go out as a **RaptorQ fountain**: a receiver that hears enough of *any* frames rebuilds the file, wherever it tuned in. No return path: bulletins and updates to a whole region.

Source: hermes-installer docs/README.md; Mercury docs/presentations/2026-10-mercury; hermes-broadcast WIRE-FORMAT.md

# hermes-radio-daemon

- **One daemon** for the radio: control, DSP, audio, voice, digital modes and the modem
- Two backends: **hfsignals** drives the sBitx and zBitx hardware itself (synthesiser, GPIO, codec); **hamlib** drives any CAT radio
- Mercury runs **inside** it, keyed and fed audio directly
- **One front door**, one HTTPS port: the web interface, REST under `/api/v2`, the `hermes.v1` WebSocket, audio over WebRTC
- Spec first: **113 operations** in an OpenAPI contract; Go handlers linked into the daemon; every route needs a login, roles enforced on the server




Source: hermes-radio-daemon README.md; docs/api-design/README.md; api/spec/api-rest.yaml

# Digital voice, and encryption where it is permitted

## Two digital voice codecs

- **RADE V2**: neural digital voice (FARGAN vocoder, the RADE V2 modem), entirely in C inside the daemon
- **D-STAR**: AMBE voice, compatible with commercial D-STAR radios
- A profile switches between analogue and digital voice

## Voice encryption (opt-in)

- For networks where encryption is permitted, **never on amateur bands**
- One 256-bit key shared by the network's stations
- D-STAR: the voice bits XORed with a ChaCha20 keystream
- RADE V2: keyed rotations of the carriers every 40 ms; a late listener locks in about 1 s

## On air, October 2026

Both codecs, encrypted and clear, with the right key, the wrong key and none, and a listener joining mid-over: all passed. Listeners heard **no difference** between encrypted and clear speech.

Source: hermes-radio-daemon README.md (Digital Voice, Voice encryption), dsp/rade_voice.h, dsp/dstar_voice.h

# Digital text modes and reception reports

- **FT8**: receive and send, a decoder with several passes and signal subtraction
- **JS8**: all four submodes (Normal, Fast, Turbo, Slow), receive and send; free-text messages across several frames
- **CW** and **RTTY**: decoded as **skimmers**, every station in the passband at once
- One log of everything decoded and sent, read through the API
- Received stations reported to **PSKReporter**, with the station's own position from its GPS
- **DRM**: digital broadcast reception, and transmission: music sent and decoded between two stations
- SSB voice processing: equaliser, compressor, limiter; on receive, adaptive noise reduction and AGC

Source: hermes-radio-daemon README.md; dsp/ft8*, dsp/js8*, dsp/pskreporter*

# Positions over the air

## Position beacons

- A station sends its position on a schedule: from its GPS, or a fixed position typed in
- Every station heard is listed with its last position, and drawn on a **map**
- The beacon format is documented byte by byte

## SOS

An emergency code set by a user goes out in every beacon, more often, until the emergency is declared over

## Tracks

- The station's own track, and every other station's from its beacons
- A point every few seconds while moving; one every 5 minutes while still
- Fixed **24-byte records**, one file per period: a time range is found by binary search
- A **trip** is a named time range of a track

Source: docs/BEACON-FORMAT.md, gps/gps_track.h, api/spec/api-rest.yaml (tag gps)

# Testing and platforms

## Testing

- **280 C unit tests** (Unity + fff): DSP, codecs, modes, beacons, tracks
- **Go**: every API handler, its refusals too
- **Integration**: the real daemon on Hamlib's dummy radio
- **Sanitizers** (ASan + UBSan, TSan) and CI on amd64 and arm64 for every pull request
- **On air**: sBitx, IC-7100 and FT-710 stations

## Platforms

- **Raspberry Pi** 4 and 5, Debian 13, from the HERMES package repository
- **A phone as the station**: the daemon and Mercury on Sailfish OS and a PinePhone Pro, the radio on USB, the phone's own GPS
- Mercury also on **Android** and **iOS** phones, **macOS** and **Windows**: the radio on USB everywhere but iOS, which gives apps no USB access
- CAT over USB without kernel drivers, proposed upstream to Hamlib

Source: tests/, .github/workflows; hermes-radio-daemon README.md; Hamlib pull request 2272

# 2026, and what comes next

**580** commits to the radio daemon since 25 April

**117** pull requests merged in it

**1468** commits to Mercury since its C rewrite

**12** stations in a network installed in November

## ⇄ Store and forward

UUCP and NNCP behind one layer of the station API; an SOS message to chosen recipients

## One interface

The web interfaces on the station API; the older WebSocket protocol retired

## Sharing the channel

HF channel access between a network's stations; Mercury's optional encryption; trips on the map

Source: git history of hermes-radio-daemon and Mercury

HERMES

# THANK YOU

github.com/Rhizomatica

hermes.radio • rhizomatica.org/hermes

lists.riseup.net/www/info/hermes-general

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